

INDIANAPOLIS INTL, IN, USA

WMO: 724380

Lat: 39.725N

Lon: 86.282W

Elev: 791

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	7.5	-6.3	4.0	2.9	-1.3	5.3	8.8	28.2	32.8	26.2	31.0	9.7	280	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	91.1	74.8	88.9	73.9	86.6	72.7	78.0	87.3	76.5	85.1	75.1	83.1	10.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	136.2	83.5	73.8	129.6	81.8	72.4	123.8	80.1	42.0	87.5	40.4	85.1	39.0	83.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.1	21.9	19.2	-4.4	94.3	8.0	3.3	-10.2	96.7	-14.8	98.6	-19.3	100.5	-25.1	102.9		
			-5.1	80.4	7.7	1.9	-10.7	81.7	-15.2	82.9	-19.5	83.9	-25.1	85.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	28.2	32.5	42.4	53.9	63.7	72.6	75.7	74.9	68.2	55.9	43.5	33.4
	DBStd	19.03	12.55	11.65	11.23	9.32	8.32	5.97	5.27	4.84	7.26	8.95	9.77	10.70
	HDD50	2346	681	499	292	64	5	0	0	0	41	241	523	
	HDD65	5224	1140	910	703	348	128	12	1	2	50	306	644	980
	CDD50	3758	7	10	57	181	430	678	797	771	546	225	48	8
	CDD65	1160	0	0	3	15	89	240	334	308	146	24	1	0
	CDH74	9670	0	0	20	117	679	1942	3014	2541	1178	177	2	0
	CDH80	3162	0	0	1	11	157	631	1103	852	373	34	0	0
Wind	WSAvg	9.5	11.0	10.9	10.9	11.3	9.5	8.4	7.7	7.3	7.7	9.2	10.1	10.2
Precipitation	PrecAvg	41.3	2.6	2.4	3.6	4.0	4.3	4.2	4.4	3.4	3.1	2.9	3.3	3.1
	PrecMax	52.6	9.4	5.4	9.0	8.6	9.3	10.3	13.1	8.3	10.4	7.8	8.5	7.7
	PrecMin	30.7	0.4	0.4	0.6	1.0	1.1	0.1	0.5	0.4	0.2	0.0	0.7	0.4
	PrecStd	6.2	1.7	1.3	1.8	1.8	2.3	2.2	2.8	2.0	2.1	1.6	1.7	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.3	67.1	76.8	81.8	87.6	91.5	96.6	93.5	91.9	84.2	72.1	64.3
		MCWB	57.9	56.6	62.3	65.5	70.6	73.5	77.3	75.0	71.4	69.3	60.3	58.6
	2%	DB	57.0	60.2	71.0	77.6	84.1	88.9	91.4	90.2	88.1	78.9	66.8	58.3
		MCWB	53.8	53.9	58.8	62.4	69.3	73.4	76.2	74.9	71.2	65.9	58.5	54.1
	5%	DB	52.0	54.9	65.4	73.6	81.3	86.4	88.7	87.5	84.3	74.6	62.8	53.6
		MCWB	49.0	49.3	55.5	60.4	67.7	72.5	74.9	73.5	69.7	63.8	56.3	50.1
	10%	DB	45.9	49.6	60.1	69.6	77.8	83.8	86.0	84.8	80.8	70.7	58.8	48.4
		MCWB	42.4	44.2	52.4	58.6	66.6	71.0	73.4	71.9	68.0	61.8	52.9	44.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	60.0	64.0	68.8	74.2	78.3	80.4	79.4	75.7	72.2	63.7	59.7
		MCDB	61.6	65.3	74.0	78.2	82.9	87.8	91.1	88.7	85.3	80.2	69.2	63.7
	2%	WB	54.6	55.1	60.5	65.2	71.8	76.3	78.5	77.4	73.8	69.1	60.3	54.9
		MCDB	56.9	58.3	68.4	72.8	81.1	85.4	87.9	86.7	83.2	75.9	64.9	57.4
	5%	WB	49.9	50.0	57.4	62.5	69.9	74.7	77.0	75.8	72.2	66.1	57.4	50.3
		MCDB	51.7	54.9	63.4	70.2	78.2	83.4	85.6	83.8	80.5	72.5	61.9	53.1
	10%	WB	42.3	44.1	53.3	60.0	68.1	73.1	75.4	74.3	70.4	62.8	53.4	45.0
		MCDB	45.5	49.1	59.3	67.9	75.6	81.0	83.1	81.5	77.5	69.0	58.4	48.0
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.6	17.7	19.6	18.3	17.5	17.6	17.7	19.3	18.5	16.0	13.4
		MCDBR	18.5	22.4	23.6	23.4	21.3	19.9	19.5	19.7	22.4	22.6	20.6	17.8
		MCWBR	16.0	17.3	14.6	13.5	10.6	8.9	7.7	7.9	8.9	11.9	14.7	15.1
	5% WB	MCDBR	17.1	20.7	21.1	19.6	18.4	17.6	17.7	17.2	18.5	18.7	18.4	16.7
		MCWBR	16.0	17.8	14.0	13.4	10.2	9.1	8.3	8.1	8.6	10.9	15.0	15.4
Clear-Sky Solar Irradiance	taub	0.303	0.323	0.348	0.404	0.438	0.442	0.457	0.453	0.418	0.359	0.321	0.301	
	taud	2.437	2.357	2.370	2.223	2.177	2.206	2.166	2.164	2.243	2.410	2.466	2.459	
	Ebn at Noon	274	283	287	276	268	266	261	259	260	267	266	266	
	Edh at Noon	26	32	35	43	47	45	47	46	40	30	25	23	
All-Sky Solar Radiation	RadAvg	559	818	1155	1505	1717	1956	1905	1739	1434	1006	672	469	
	RadStd	59	93	100	122	156	118	112	101	117	97	74	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.85	N/A	N/A	+182	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	N/A

Nomenclature: See separate page